Integration by Substitution

Questions 1–66

Question 1. Each of the following integrals can be found using reversing the chain rule. However, in each case, evaluate the integrals using the given substitution:

a) $\int (3x+1)^4 \, \mathrm{d}x$ $u = 3x + 1$

b) $\int \frac{4}{(2x-3)^3} \, \mathrm{d}x$ $u = 2x - 3$

c) $\int \sqrt{8x+5} \, \mathrm{d}x$ $u = 8x + 5$

d) $\int \frac{6}{\sqrt[3]{5-2x}} \, \mathrm{d}x$ $u = 5 - 2x$

Question 2. It is given that $y = x(x-3)^5$

a) Describe the transformation that would map $y = x(x-3)^5$ onto the image $y = (x+3)x^5$

b) Use the substitution $u = x - 3$ to show that

$$\int x(x-3)^5 \, \mathrm{d}x = \int (u+3)u^5 \, \mathrm{d}u$$

Question 3. It is given that $y = 2x\sqrt{x+4}$

a) Describe the transformation that would map $y = 2x\sqrt{x+4}$ onto the image $y = 2(x-4)\sqrt{x}$

b) Use the substitution $u = x + 4$ to show that

$$\int 2x\sqrt{x+4}\,\mathrm{d}x = \int 2(u-4)\sqrt{u}\,\mathrm{d}u$$

Question 4. Evaluate the following integrals using the given substitution:

a) $\int x\sqrt{x+1}\,\mathrm{d}x$ $u = x + 1$

b) $\int 2x\sqrt{2x+3}\,\mathrm{d}x$ $u = 2x + 3$

c) $\int x^3\sqrt{x^2+1}\,\mathrm{d}x$ $u = x^2 + 1$

Question 5. Evaluate the following integrals using the given substitution:

a) $\int \frac{x}{x+1}\,\mathrm{d}x$ $u = x + 1$

b) $\int \frac{2x}{2x+3}\,\mathrm{d}x$ $u = 2x + 3$

c) $\int \frac{x^3}{x^2+1}\,\mathrm{d}x$ $u = x^2 + 1$

Question 6. Evaluate the following integrals using the given substitution:

a) $\int \frac{x^2}{x+1}\,\mathrm{d}x$ $u = x + 1$

b) $\int \frac{4x^2}{2x+3}\,\mathrm{d}x$ $u = 2x + 3$

c) $\int \frac{x^5}{x^2+1}\,\mathrm{d}x$ $u = x^2 + 1$

Question 7. Evaluate the following integrals using the given substitution:

a) $\int x^2\sqrt{x+1}\,\mathrm{d}x$ $u = x+1$

b) $\int 4x^2\sqrt{2x+3}\,\mathrm{d}x$ $u = 2x+3$

c) $\int x^3\sqrt{x^2+1}\,\mathrm{d}x$ $u = x^2+1$

Question 8. Use the substitution $u = 2-3x$ to find

$$\int 189x(2-3x)^5\,\mathrm{d}x$$

Question 9. Use the substitution $u = x + 1$ to find the exact value of

$$\int_0^1 3x(x+1)^{-\frac{1}{2}}\,\mathrm{d}x$$

Question 10. Use the substitution $u = 3x + 4$ to show that

$$\int_{-2}^{-\frac{4}{3}} (6x+1)(3x+4)^7\,\mathrm{d}x = \frac{3040}{27}$$

Question 11. Use integration by substitution to show that

$$\int_1^2 \frac{x}{x+2}\,\mathrm{d}x = a + \ln\frac{b}{c}$$

where a, b and c are positive integers.

Question 12. Use integration by substitution to show that

$$\int_1^3 \frac{9x}{3x+5}\,\mathrm{d}x = 6 + 5\ln\frac{4}{7}$$

Question 13. Use the substitution $u = 2x + 3$ to find the exact value of

$$\int_1^2 \frac{8x - 1}{(2x + 3)^3} \, \mathrm{d}x$$

Question 14. Use integration by substitution to show that

$$\int_{\frac{3}{2}}^{\frac{19}{2}} 5x\sqrt{2x - 3} \, \mathrm{d}x = 672$$

Question 15. Use integration by substitution to show that

$$\int_0^1 140x(4x+1)^{\frac{3}{2}}\,\mathrm{d}x = 1 + 225\sqrt{5}$$

Question 16. Use integration by substitution to find

$$\int (30x-45)\sqrt{x+2}\,\mathrm{d}x$$

in the form $(x+2)^{\frac{3}{2}}(ax+b)+c$, where a and b are integers, and c is a constant.

Question 17. Use integration by substitution to find the exact value of

$$\int_{\frac{1}{12}}^{\frac{3}{4}} (2-6x)(12x-1)^{\frac{1}{3}}\,\mathrm{d}x$$

Question 18. Use the substitution $u = x^2 + 5$ to find

$$\int \frac{2x^3 - x}{\sqrt{x^2+5}}\,\mathrm{d}x$$

Question 19. Use the substitution $u = 4 - x^4$ to find the exact value of

$$\int_0^1 \frac{x^7}{4 - x^4}\,\mathrm{d}x$$

Question 20. Use the substitution $u = x^5 + 3$ to show that

$$\int_0^1 \frac{20x^9}{(x^5 + 3)^2}\,\mathrm{d}x = \ln\frac{a}{b} - 1$$

where a and b are positive integers.

Question 21. Use the substitution $u = 2\sqrt{x} - 1$ to find

$$\int \frac{2}{2\sqrt{x} - 1} \, \mathrm{d}x$$

Question 22. Evaluate the following definite integral

$$\int_1^4 \frac{2}{\sqrt{x} + 1} \, \mathrm{d}x$$

using the substitution $u = \sqrt{x} + 1$

Question 23. Evaluate the following definite integral

$$\int_2^5 \frac{2x^2}{\sqrt{x-1}}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x-1}$

Question 24. Use integration by substitution to find

$$\int \frac{3x}{1+\sqrt{x}}\,\mathrm{d}x$$

Question 25. Evaluate the following definite integral

$$\int_1^4 \frac{1}{\sqrt{x}(2+\sqrt{x})}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x}$

Question 26. Use the substitution $u = 1 + 2\sqrt{x}$ to show that

$$\int_1^9 \frac{1+\sqrt{x}}{1+2\sqrt{x}}\,\mathrm{d}x$$

can be written in the form $a + b\ln c$, where a is an integer, and b and c are rational numbers.

Question 27. Evaluate the following definite integral

$$\int_0^3 \frac{8-x}{3+\sqrt{x+1}}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x+1}$

Question 28. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{16} \frac{4}{1+\sqrt{x}}\,\mathrm{d}x$ $u^2 = x$

b) $\int_1^3 \frac{8x}{\sqrt{4x-3}}\,\mathrm{d}x$ $u^2 = 4x - 3$

Question 29. Evaluate the following definite integral:

$$\int_1^4 \frac{12}{4\sqrt{x}+3x}\,\mathrm{d}x$$

a) Using the substitution $u = \sqrt{x}$
b) Using the substitution $u = 3\sqrt{x}+4$

Question 30. Use the substitution $u^2 = 2x^2+1$ to evaluate

$$\lim_{\delta x \to 0} \sum_{x=0}^{2} \frac{x^3}{\sqrt{2x^2+1}} \delta x$$

Question 31. Standard Functions Each of the following integrals can be found using reversing the chain rule. However, in each case, evaluate the integrals using the given substitution:

a) $\int \cos x \sin^4 x \, \mathrm{d}x$ $u = \sin x$

b) $\int 8xe^{x^2+1} \, \mathrm{d}x$ $u = x^2 + 1$

c) $\int \sin x \, e^{\cos x} \, \mathrm{d}x$ $u = \cos x$

d) $\int \frac{\sec^2 x}{\tan x} \, \mathrm{d}x$ $u = \tan x$

Question 32. Evaluate the following definite integral

$$\int_0^{\frac{\pi}{2}} \frac{6\cos x}{9 + \sin x} \, \mathrm{d}x$$

using the substitution $u = 9 + \sin x$

Question 33. Use integration by substitution to find

$$\int \frac{5\mathrm{e}^x}{5+\mathrm{e}^x}\,\mathrm{d}x$$

Question 34. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\ln 2} \frac{3\mathrm{e}^{2x}}{3+\mathrm{e}^x}\,\mathrm{d}x$ $u = 3+\mathrm{e}^x$

b) $\int_{\ln 2}^{\ln 3} \frac{8\mathrm{e}^x}{(2+\mathrm{e}^x)^2}\,\mathrm{d}x$ $u = 2+\mathrm{e}^x$

Question 35. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\frac{\pi}{4}} 8\cos 2x \sin^5 2x \, \mathrm{d}x$ $u = \sin 2x$

b) $\int_0^{\frac{\pi}{8}} \sin^2 2x \cos^3 2x \, \mathrm{d}x$ $u = \sin 2x$

Question 36. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\frac{\pi}{4}} \sec^2 x \, \mathrm{e}^{\tan x + 1} \, \mathrm{d}x$ $u = \tan x + 1$

b) $\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt[3]{\tan x} \, \mathrm{d}x$ $u = \tan x$

Question 37. Use the substitution $u = 1 + \tan x$ to find

$$\int \frac{1}{\cos^2 x(1+\tan x)^2}\,\mathrm{d}x$$

Question 38. Evaluate the following definite integral

$$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x$$

a) Using the substitution $u = 5^x - 1$
b) Using the substitution $u = 5^x$

Question 39. Evaluate the following definite integral

$$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x$$

a) Using the substitution $u = 8^x + 1$
b) Using the substitution $u = 8^x$

Question 40. Use the substitution $u = 3 - \ln x$ to find

$$\int \frac{6}{x(3-\ln x)^5}\,\mathrm{d}x$$

Question 41. Evaluate the following definite integrals using the given substitutions:

a) $\int_1^{2e} \frac{4}{x(5+\ln x)^3}\,\mathrm{d}x$ $u = 5 + \ln x$

b) $\int_1^{e} \frac{2\ln x}{x(4-\ln x)^2}\,\mathrm{d}x$ $u = 4 - \ln x$

Question 42. Trigonometric and other substitutions Use the substitution $x = \sin u$ to find:

a) $\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

b) $\int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,\mathrm{d}x$

Question 43. Evaluate the following definite integral

$$\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x$$

Question 44. Use the substitution $x = \frac{3}{2}\sin u$ to find

$$\int \frac{1}{\sqrt{9-4x^2}}\,\mathrm{d}x$$

Question 45. Use the given substitutions to find:

a) $\int \frac{1}{1+x^2}\,\mathrm{d}x$ $x = \tan u$

b) $\int \frac{5}{25+x^2}\,\mathrm{d}x$ $x = 5\tan u$

Question 46. Evaluate the following definite integral

$$\int_0^1 \frac{1}{(1+x^2)^2}\,\mathrm{d}x$$

using the substitution $x = \tan u$

Question 47. Evaluate the following definite integral

$$\int_0^1 \frac{1-x^2}{1+x^2}\,\mathrm{d}x$$

using the substitution $x = \tan u$

Question 48. Evaluate the following definite integrals using the given substitutions:

a) $\int_2^3 \frac{1}{(x-2)^2+1}\,\mathrm{d}x$ $x = 2 + \tan u$

b) $\int_{-3}^{-1} \frac{1}{(x+3)^2+4}\,\mathrm{d}x$ $x = 2\tan u - 3$

Question 49. Use integration by substitution to find

$$\int_5^8 \frac{1}{x^2 - 10x + 34}\,\mathrm{d}x$$

Question 50. Use integration by substitution to find

$$\int_5^8 \frac{12x}{x^2 - 10x + 34}\,\mathrm{d}x$$

Question 51. Evaluate the following definite integral

$$\int_{\frac{2\sqrt{3}}{3}}^{2} \frac{6}{x^2\sqrt{x^2-1}}\,\mathrm{d}x$$

using the substitution $x = \operatorname{cosec} u$

Question 52. Given that $x > 3$, use the substitution $x = 3\operatorname{cosec} u$ to show that

$$\int \frac{1}{x^2\sqrt{x^2-9}}\,\mathrm{d}x = \frac{\sqrt{x^2-9}}{9x} + c$$

Question 53. Prerequisite knowledge for Q53: Integration by Parts Use the substitution $2x = \mathrm{e}^u$ to find

$$\int (\ln 2x)^2 \, \mathrm{d}x$$

Question 54. Use the substitution $x = \sin u$ to find

$$\int \frac{3}{(1-x^2)^{\frac{5}{2}}} \, \mathrm{d}x$$

Question 55. Prerequisite knowledge for Q55–Q58: Double Angle Formulae Use the substitution $u = \cos x - 2$ to find

$$\int \frac{4\sin 2x}{\cos x - 2}\,\mathrm{d}x$$

Question 56. Use integration by substitution to find

$$\int \frac{x^2}{\sqrt{1-x^2}}\,\mathrm{d}x$$

Question 57. Evaluate the following definite integral using the given substitution:

a) $\int_0^{\frac{3}{2}} \sqrt{9-x^2}\,\mathrm{d}x$ $x = 3\sin u$

b) $\int_0^{\frac{1}{3}} \sqrt{1-9x^2}\,\mathrm{d}x$ $x = \frac{1}{3}\sin u$

Question 58. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^1 \frac{1-x^2}{(1+x^2)^2}\,\mathrm{d}x$ $x = \tan u$

b) $\int_0^{\frac{5}{2}} \sqrt{\frac{x}{5-x}}\,\mathrm{d}x$ $x = 5\sin^2 u$

Question 59. Prerequisite knowledge for Q59–Q62: Integration by Parts Use integration by substitution to find:

a) $\int 2x^3 \ln(x^2+4)\,\mathrm{d}x$

b) $\int 3^x \ln(3^x+1)\,\mathrm{d}x$

Question 60. Use the substitution $u = \sqrt{x+2}$ to find

$$\int \mathrm{e}^{4\sqrt{x+2}}\,\mathrm{d}x$$

Question 61. Evaluate the following definite integral

$$\int_0^4 \mathrm{e}^{\sqrt{2x+1}}\,\mathrm{d}x$$

using the substitution $u^2 = 2x + 1$

Question 62. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^1 5x^5 \ln(10x^3+1)\,\mathrm{d}x$ $u = 10x^3 + 1$

b) $\int_{-1}^{\frac{\pi^2}{4}-1} \cos(\sqrt{x+1})\,\mathrm{d}x$ $u = \sqrt{x+1}$

Question 63. Prerequisite knowledge for Q63–Q65: Partial Fractions Evaluate the following definite integrals

$$\int_0^{\ln 5} \frac{10}{5+\mathrm{e}^x}\,\mathrm{d}x$$

using the substitution $u = \mathrm{e}^x$

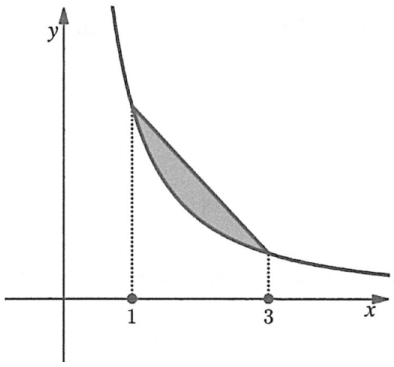
Question 64. Evaluate the following definite integrals

$$\int_1^4 \frac{2}{x(1+3\sqrt{x})}\,\mathrm{d}x$$

using the substitution $x = u^2$

Question 65. The diagram below shows the curve with equation

$$y = \frac{1}{x\sqrt{x+1}}$$



a) Using the substitution $u = \sqrt{x+1}$, find the exact value of the definite integral

$$\int_1^3 \frac{1}{x\sqrt{x+1}}\,\mathrm{d}x$$

b) Find the exact value of the shaded area.

Question 66. Prerequisite knowledge for Q66: Implicit Differentiation

a) Use the substitution $\mathrm{e}^x = \sec^2 u$ to show that:

$$\int \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x = \int 2\tan^2 u\,\mathrm{d}u$$

b) Hence find the exact value of:

$$\int_0^{\ln\frac{4}{3}} \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x$$